

Quantum Mechanics II Exccercise 1

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1 Natrual Units

1.1 Unit Equivilence

In CGS, the speed of light is measured in cm/sec . Therefore, for a unit system where $c = 1$, we get:

$$c = 3 \cdot 10^{10} cm/sec = 1 \quad (1)$$

$$1 sec = 3 \cdot 10^{10} cm \quad (2)$$

Therefore, distance and time are measured using the same units. Using the same logic, in CGS, mass is measured in grams and energy is measured in ergs. Assigning equation (2) into the definition of 1 erg, we get:

$$1 erg = 1 g \cdot \frac{cm^2}{sec^2} = 1 g \cdot \frac{cm^2}{9 \cdot 10^{20} cm^2} = \frac{1}{9} \cdot 10^{-20} g \quad (3)$$

And so we got that, assuming $c = 1$, the units for mass and energy are unified. This can also be shown using the relation from special relativity $E = mc^2$. As instructed, the units of mass and energy will be eV .

Now let's use the fact that $\hbar = 1$. $\hbar$ Has units of angualr momentum. Angular momentum has units of $energy \times time$. Since $\hbar$ is unitless, this means that angular momentum is unitless, and requires the units of energy to be opposite to the unit of time:

$$\hbar = 10^{-27} erg \cdot sec = 1 \quad (4)$$

$$1 sec = 10^{27} erg^{-1} \quad (5)$$

Therefore the units of distance and time are opposite to the unit of energy and mass, and eV^{-1} can be used to measure them. Writing all of the units in eV is now just an issue of conversion from erg.

$$1 erg = 6.242 \cdot 10^{11} eV \quad (6)$$

$$1 g = 9 \cdot 10^{20} erg = 5.617 \cdot 10^{32} eV \quad (7)$$

$$1 sec = 10^{27} erg^{-1} = 1.6 \cdot 10^{15} eV^{-1} \quad (8)$$

$$1 cm = sec/(3 \cdot 10^{10}) = 5.34 \cdot 10^4 eV^{-1} \quad (9)$$

1.2 Newton's Constant and Plank Mass

Using the values from equations (6) to (9):

$$G_N = 6.7 \cdot 10^{-8} cm^3 sec^{-2} g^{-1} \quad (10)$$

$$G_N = 6.7 \cdot 10^{-8} \cdot (5.34 \cdot 10^4 eV^{-1})^3 \cdot (1.6 \cdot 10^{15} eV^{-1})^{-2} \cdot (5.617 \cdot 10^{32} eV)^{-1} \quad (11)$$

$$G_N = 7.095 \cdot 10^{-57} eV^{-2} \quad (12)$$

This is the very close to the value we would get using Plank Mass $M_{pl} = 1.2 \cdot 10^{19} GeV$:

$$G_N = M_{pl}^{-2} = (1.2 \cdot 10^{19} GeV)^{-2} = 6.94 \cdot 10^{-57} eV^{-2} \quad (13)$$

With a difference of around 2.1%, resulting probably from our rounding of the numbers.

1.3 The Actions of different systems

1.3.1 Tennis ball in a court

The average serve velocity is around $174 \text{ km/h} = 48.33 \text{ m/s}$ ¹. The length of a tennis court is 23.78 meters. Assuming the ball is being served horizontally, the time it would take to complete that distance is 0.492 seconds. In that time, the ball drops

$$\Delta y = \frac{1}{2}gt^2 = 1.18 \text{ m} \quad (14)$$

Assuming the pitch is done from an average height of 1.5 meters, we can ignore the floor. We will also ignore air resistance, even though it affects the motion, it will not make a visible change in the action relative to that of the water molecule. The speed of the ball is then, as a function of time:

$$\dot{x}^2 = v_0^2 + (gt)^2 = 2335.79 + 96.04t^2 \quad (15)$$

$$\begin{aligned} & \frac{1}{\hbar} S[x_c(t)] = \\ & = \frac{1}{\hbar} \int_0^{0.492 \text{ sec}} \frac{1}{2} 0.056 \text{ kg} \cdot (2335.79 + 96.04t^2) dt \sim 3.2 \cdot 10^{35} \end{aligned} \quad (16)$$

1.3.2 Water molecule in a teacup

The mean free path of a water molecule is roughly $2.5 \text{ \AA}$ ². During this time, the molecule moves in an average velocity depending on the temperature, following the formula

$$v = \sqrt{\frac{2k_B T}{m}} \quad (17)$$

Assuming the temperature of the tea is roughly 60°C and the mass of a water molecule is $3 \cdot 10^{-26} \text{ kg}$, we get an average speed of 553.75 m/s . This makes the mean free time $T = 4.5 \cdot 10^{-13}$ seconds. Assigning all of these into the action formula, we get:

$$\begin{aligned} & \frac{1}{\hbar} S[x_c(t)] = \\ & = \frac{1}{\hbar} \int_0^{4.5 \cdot 10^{-13}} \frac{1}{2} 10^{-26} \text{ kg} \cdot (553.75 \text{ m/s})^2 dt \sim 21 \end{aligned} \quad (18)$$

2 Smeared Amplitude

2.1 $K(\vec{x}, t)$ is a distribution: Intuitive explanation

In the integration over d^3x , the r in the denominator would cancel out with the r in the Jacobian. As for the integrand, the rapid oscillations near its singularities could be canceled by a bounded test function, resulting in a finite numerical value.

2.2 Smeared Amplitude of a 3D Gaussian

$$\overline{K}(\vec{X}, t) = \int d^3x \left(\frac{1}{\pi l^2} \right)^{3/2} e^{-\left(\frac{\vec{x}-\vec{X}}{l} \right)^2} \frac{-i}{(2\pi)^2 r} \int_{-\infty}^{\infty} dp p e^{i(pr - t\sqrt{m^2 + p^2})} \quad (19)$$

¹<https://toomanyrackets.com/average-serve-speeds-in-tennis/>

²<https://galileo.phys.virginia.edu/classes/304/h2o.pdf>

Without limitation of generality, we can set our coordinate system such that X is on the z axis for the sake of integration. This means that

$$(\vec{x} - \vec{X})^2 = r^2 + R^2 - 2\vec{x} \cdot \vec{X} = r^2 + R^2 - 2rR \cos(\theta) \quad (20)$$

Where $R = |\vec{X}|$ and $\vec{x} = (r, \theta, \varphi)$ in spherical coordinates. This makes our integral into

$$\begin{aligned} \bar{K}(\vec{X}, t) &= \int d^3x \left(\frac{1}{\pi l^2} \right)^{3/2} e^{-\frac{r^2 + R^2 - 2rR \cos(\theta)}{l^2}} \frac{-i}{(2\pi)^2 r} \int_0^\infty dp p e^{i(pr - t\sqrt{m^2 + p^2})} = \\ &= \left(\frac{1}{\pi l^2} \right)^{3/2} \frac{-i}{(2\pi)^2} \int_0^\infty dp p e^{-it\sqrt{m^2 + p^2}} \int_0^\infty dr \int_0^\pi d\theta \int_0^{2\pi} d\varphi r^2 \sin(\theta) \frac{1}{r} \exp \left[-\frac{r^2 + R^2 - 2rR \cos(\theta)}{l^2} + ipr \right] \end{aligned} \quad (21)$$

The integral over φ gives 2π , while the integrals over r and θ are gaussians. Starting from the integral over r we perform a variable change

$$\begin{aligned} u &= \frac{r - v}{l} \quad ; \quad v = R \cos(\theta) + \frac{ipl^2}{2} \\ u^2 &= \frac{1}{l^2} \left[r^2 + \left(R \cos(\theta) + \frac{ipl^2}{2} \right)^2 - 2Rr \cos(\theta) \right] - ipr \\ r &= lu + v \quad ; \quad dr = ldu \end{aligned} \quad (22)$$

Which gives the r integral

$$\begin{aligned} \int_0^\infty dr r \exp \left[-\frac{r^2 + R^2 - 2rR \cos(\theta)}{l^2} + ipr \right] &= \\ = l \exp \left[-\frac{R^2}{l^2} + \frac{R \cos(\theta) + \frac{ipl^2}{2}}{l^2} \right] \int_0^\infty du (lu + v) e^{-u^2} \end{aligned} \quad (23)$$

The integral in the last part evaluates to

$$\begin{aligned} \int_0^\infty du (lu + v) e^{-u^2} &= \\ = l \int_0^\infty u e^{-u^2} du + v \int_0^\infty e^{-u^2} du &= \\ = -\frac{l}{2} [e^{-u^2}]_0^\infty + \frac{\sqrt{\pi}v}{2} = \frac{1}{2} [\sqrt{\pi}v + l] \end{aligned} \quad (24)$$

Assigning it back into equation 21 using the expression from the last part of equation 23, and assigning the integral over φ to be 2π , we get:

$$\begin{aligned} \bar{K}(\vec{X}, t) &= \\ = \left(\frac{1}{\pi l^2} \right)^{3/2} \frac{-i}{2\pi} \int_0^\infty dp p e^{-it\sqrt{m^2 + p^2}} \int_0^\pi d\theta \sin(\theta) l \exp \left[-\frac{R^2}{l^2} + \frac{R \cos(\theta) + \frac{ipl^2}{2}}{l^2} \right] \frac{1}{2} \left[\sqrt{\pi} \left(R \cos(\theta) + \frac{ipl^2}{2} \right) + l \right] \end{aligned} \quad (25)$$

Moving to integral over v as it is defined in equation 22, and noticing that $\sin(\theta)d\theta = dv/R$:

$$\begin{aligned} \bar{K}(\vec{X}, t) &= \\ = \left(\frac{1}{\pi l^2} \right)^{3/2} \frac{-il}{2\pi R} e^{-\frac{R^2}{l^2}} \int_0^\infty dp p e^{-it\sqrt{m^2 + p^2}} \int_{v+}^{v-} dv \frac{1}{2} [\sqrt{\pi}v + l] e^{\frac{v^2}{l^2}} \end{aligned} \quad (26)$$

Where $v_{\pm} = v(\cos(\theta) = \pm 1) = \pm R + \frac{ipl^2}{2}$ Which again gives the complex gaussian integral, over the variable iv this time, and with finite limits. The integral of the complex gaussian function over a subsection of the plane is known as the imaginary error function **erfi**.

$$\int_{v_+}^{v_-} dv \frac{1}{2} [\sqrt{\pi}v + l] e^{\frac{v^2}{l^2}} = \frac{1}{2} \left[\sqrt{\pi}l^2 \left(e^{\frac{v_-^2}{l^2}} - e^{\frac{v_+^2}{l^2}} \right) + l^2 \left(\operatorname{erfi} \left(\frac{v_-}{l} \right) - \operatorname{erfi} \left(\frac{v_+}{l} \right) \right) \right] \quad (27)$$

And therefore

$$\begin{aligned} \bar{K}(\vec{X}, t) &= \left(\frac{1}{\pi l^2} \right)^{3/2} \frac{-il}{2\pi R} e^{-\frac{R^2}{l^2}} \int_0^\infty dp p e^{-it\sqrt{m^2+p^2}} \frac{1}{2} \left[\sqrt{\pi}l^2 \left(e^{\frac{v_-^2}{l^2}} - e^{\frac{v_+^2}{l^2}} \right) + l^2 \left(\operatorname{erfi} \left(\frac{v_-}{l} \right) - \operatorname{erfi} \left(\frac{v_+}{l} \right) \right) \right] = \\ &= \frac{1}{2\pi^2 R} \int_0^\infty dp p e^{-it\sqrt{m^2+p^2}} \left[e^{-\frac{(pl)^2}{4}} \sin(pR) + \frac{i}{2\sqrt{\pi}} e^{-\frac{R^2}{l^2}} \left(\operatorname{erfi} \left(\frac{v_-}{l} \right) - \operatorname{erfi} \left(\frac{v_+}{l} \right) \right) \right] \end{aligned} \quad (28)$$

Using the definition of $\operatorname{erfi} = -ierf(ix)$, we get the final value

$$\bar{K}(\vec{X}, t) = \frac{1}{2\pi^2 R} \int_0^\infty dp p e^{-it\sqrt{m^2+p^2}} \left[e^{-\frac{(pl)^2}{4}} \sin(pR) + \frac{1}{2\sqrt{\pi}} e^{-\frac{R^2}{l^2}} \left(\operatorname{erf} \left(\frac{R}{l} - \frac{ipl}{2} \right) - \operatorname{erf} \left(-\frac{R}{l} - \frac{ipl}{2} \right) \right) \right] \quad (29)$$

If we take the limit $R \gg l$, the error functions go to ± 1 at infinity, but the exponential factor in their terms goes to zero, and so that whole term disappears and we are left with

$$\bar{K}(\vec{X}, t)_{R \gg l} = \frac{1}{2\pi^2 R} \int_0^\infty dp p e^{-it\sqrt{m^2+p^2}} e^{-\frac{(pl)^2}{4}} \sin(pR) \quad (30)$$

Opening up the sine will give the same expression as seen in the exercise, but twice, with opposite signs. Switching the sign of the second term will give us the same expression as in the exercise.

$$\bar{K}(\vec{X}, t)_{R \gg l} = \frac{1}{2\pi^2 R} \int_0^\infty dp p e^{-\frac{(pl)^2}{4}} e^{i(pR - t\sqrt{m^2+p^2})} \quad (31)$$

2.3 Closed-Form Estimate of $\bar{K}(\vec{X}, t)$

To integrate over p , we must use a contour Couchy integration without passing through areas where the function is singular. The damping term $e^{-\frac{(pl)^2}{4}}$ diverges when $-\operatorname{Re}\{p^2\} = |p|^2(\sin^2(\arg(p)) - \cos^2(\arg(p))) > 0$ meaning which happens in the top and bottom quarters of the plane $|\cos(\arg(p))| > |\sin(\arg(p))|$. This, of course, is only a problem if p goes to infinity in this region. If we limit it to go to infinity only on the real line, the problem would vanish.

Meanwhile, using the approximation $R \gg t$, the saddle point is $p = \pm im$. In order to work with both of these limitations, we will choose a contour that starts from $-\infty$, goes to $-\varepsilon$ where $\varepsilon \rightarrow 0$ and $\varepsilon > 0$, then goes up along the imaginary axis with values $-\varepsilon + iy$ until it passes through the saddle point, then goes down along the line $+\varepsilon + iy$ until it reaches the real line, and from there to infinity. This still leaves the problem of the singularity of the exponent at the saddle point. To avoid this problem we will use a Taylor approximation to second order $pR - t\sqrt{m^2+p^2} \approx pr - \frac{it}{2}(m^2 + p^2)$. Now we are ready to use the saddle point approximation, we start by calculating the second derivative of the exponent at the point and taking its argument $\theta = \arg(-it) = 1.5\pi$. This gives the value of the integral

$$I_C = ime^{\frac{t^2 m^2}{4}} e^{-i\pi/4} \sqrt{\frac{2\pi}{Rt}} = e^{i\pi/4} me^{\frac{t^2 m^2}{4}} \sqrt{\frac{2\pi}{Rt}} \quad (32)$$

3 Discrete version of the free scalar field

3.1 Commutation relations based on the Fourier modes

$$\begin{aligned}
[\phi_n, \phi_m] &= \left[\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{ink} \phi(k), \int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{imp} \phi(p) \right] = \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{i(nk+mp)} [\phi(k), \phi(p)] = 0 \\
[\pi_n, \pi_m] &= \left[\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{ink} \pi(k), \int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{imp} \pi(p) \right] = \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{i(nk+mp)} [\pi(k), \pi(p)] = 0 \\
[\phi_n, \pi_m] &= \left[\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{ink} \phi(k), \int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{imp} \pi(p) \right] = \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{i(nk+mp)} [\phi(k), \pi(p)] = \\
&= \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{i(nk+mp)} i \sum_{l=-\infty}^{\infty} e^{-il(k+p)} = \\
&= i \sum_{l=-\infty}^{\infty} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{i(nk+mp)-il(k+p)} = \\
&= i \sum_{l=-\infty}^{\infty} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{i(n-l)k} e^{i(m-l)p} = \\
&= i \sum_{l=-\infty}^{\infty} \delta_{nl} \delta_{ml} = i \delta_{nm}
\end{aligned} \tag{33}$$

3.2 The Hamiltonian based on the Fourier modes

$$\begin{aligned}
H &= \frac{1}{2} \sum_{n=-\infty}^{\infty} \left[\pi_n^2 + (\phi_{n+1} - \phi_n)^2 + m^2 \phi_n^2 \right] = \\
&= \frac{1}{2} \sum_{n=-\infty}^{\infty} \left[\left(\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{ink} \pi(k) \right)^2 + \left(\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{i(n+1)k} \phi(k) - \int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{ink} \phi(k) \right)^2 + m^2 \left(\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{ink} \phi(k) \right)^2 \right]
\end{aligned} \tag{34}$$

The momentum dependent term of the Hamiltonian can be simplified using the substitution $p \rightarrow -p$ in the second integral, inverting the integration order and using the relations $\pi(-p) = \pi^\dagger(p)$ and $\sum_n e^{i(k-p)n} = 2\pi\delta(k-p)$:

$$\begin{aligned}
H_p &= \frac{1}{2} \sum_{n=-\infty}^{\infty} \left(\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{ink} \pi(k) \right)^2 = \frac{1}{2} \sum_{n=-\infty}^{\infty} \left(\int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{inp} \pi(p) \right) \left(\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{ink} \pi(k) \right) = \\
&= \frac{1}{2} \sum_{n=-\infty}^{\infty} \left(\int_{+\pi}^{-\pi} \frac{-dk}{2\pi} e^{-ink} \pi(-k) \right) \left(\int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{inp} \pi(p) \right) = \\
&= \frac{1}{2} \sum_{n=-\infty}^{\infty} \left(\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{-ink} \pi^\dagger(k) \right) \left(\int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{inp} \pi(p) \right) = \\
&= \frac{1}{2} \sum_{n=-\infty}^{\infty} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{-in(k-p)} \pi^\dagger(k) \pi(p) = \\
&= \frac{1}{2} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \int_{-\pi}^{+\pi} \frac{dp}{2\pi} \pi^\dagger(k) \pi(p) \sum_{n=-\infty}^{\infty} e^{-in(k-p)} = \\
&= \frac{1}{2} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \int_{-\pi}^{+\pi} \frac{dp}{2\pi} \pi^\dagger(k) \pi(p) 2\pi\delta(k-p) = \frac{1}{2} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \pi^\dagger(k) \pi(k)
\end{aligned} \tag{35}$$

The term of the restoring force follows the exact same logic:

$$\frac{1}{2} \sum_{n=-\infty}^{\infty} m^2 \left(\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{ink} \phi(k) \right)^2 = \frac{1}{2} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} m^2 \phi^\dagger(k) \phi(k) \quad (36)$$

And the interaction term is

$$\begin{aligned} & \frac{1}{2} \sum_{n=-\infty}^{\infty} \left(\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{i(n+1)k} \phi(k) - \int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{ink} \phi(k) \right)^2 = \\ &= \frac{1}{2} \sum_{n=-\infty}^{\infty} \left(\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{ink} \phi(k) (e^{ik} - 1) \right)^2 = \\ &= \frac{1}{2} \sum_{n=-\infty}^{\infty} \left(\int_{-\pi}^{+\pi} \frac{dk}{2\pi} e^{ink} \phi(k) (e^{ik} - 1) \right) \left(\int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{inp} \phi(p) (e^{ip} - 1) \right) = \\ &= \frac{1}{2} \sum_{n=-\infty}^{\infty} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \int_{-\pi}^{+\pi} \frac{dp}{2\pi} e^{-in(k-p)} \phi^\dagger(k) \phi(p) (e^{-ik} - 1) (e^{ip} - 1) = \\ &= \frac{1}{2} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \phi^\dagger(k) \phi(k) (e^{-ik} - 1) (e^{ik} - 1) = \\ &= \frac{1}{2} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} \phi^\dagger(k) \phi(k) (2 - 2 \cos(k)) = \end{aligned} \quad (37)$$

Putting it all together we get

$$\begin{aligned} H &= \frac{1}{2} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} [\pi^\dagger(k) \pi(k) + (m^2 + 2(1 - \cos(k))) \phi^\dagger(k) \phi(k)] = \\ &= \frac{1}{2} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} [\pi^\dagger(k) \pi(k) + \omega_k^2 \phi^\dagger(k) \phi(k)] \end{aligned} \quad (38)$$

With the dispersion relation

$$\omega_k = \sqrt{m^2 + 2(1 - \cos(k))} \quad (39)$$

With the lattice spacing Δ , it becomes

$$\omega_k = \sqrt{m^2 + \frac{2}{\Delta^2} (1 - \cos(k\Delta))} \quad (40)$$

And at the limit $k\Delta \ll 1$ we get

$$\frac{2}{\Delta^2} (1 - \cos(k\Delta)) = \frac{2}{\Delta^2} \left[1 - \left(1 - \frac{(k\Delta)^2}{2} \right) \right] = \frac{(k\Delta)^2}{\Delta^2} = k^2 \quad (41)$$

Which is the Klein-Gordon dispersion relation from Special Relativity:

$$\omega_k = \sqrt{m^2 + k^2} \quad (42)$$

3.3 Ladder Operators

$$a_k = \frac{1}{\sqrt{4\pi\omega_k}} (\omega_k \phi(k) + i\pi(k)), \quad a_k^\dagger = \frac{1}{\sqrt{4\pi\omega_k}} (\omega_k \phi^\dagger(k) - i\pi^\dagger(k)) \quad (43)$$

The commutation relations are

$$\begin{aligned}
[a_k, a_p] &= \frac{1}{4\pi\sqrt{\omega_k\omega_p}} [\omega_k\phi(k) + i\pi(k), \omega_p\phi(p) + i\pi(p)] = \\
&= \frac{1}{4\pi\sqrt{\omega_k\omega_p}} [\omega_k\omega_p[\phi(k), \phi(p)] + i(\omega_k[\phi(k), \pi(p)] + \omega_p[\pi(k), \phi(p)]) - [\pi(k), \pi(p)]] = \\
&= -\frac{1}{4\pi\sqrt{\omega_k\omega_p}} \left(\omega_k \sum_{n=-\infty}^{\infty} e^{-in(k+p)} - \omega_p \sum_{n=-\infty}^{\infty} e^{-in(k+p)} \right) = \\
&= -\frac{\omega_k - \omega_p}{4\pi\sqrt{\omega_k\omega_p}} 2\pi\delta(k+p) = 0
\end{aligned} \tag{44}$$

Where in the last part the delta function forced $k = -p$, in which case $\omega_k = \omega_p$ since ω is invariant to the sign of k . From this commutation relations the commutation relations $[a_k^\dagger, a_p^\dagger]$ are obvious to develop using the relations $\phi^\dagger(k) = \phi(-k)$ and $\pi^\dagger(k) = \pi(-k)$, and a variable substitution back $k \rightarrow -k'$ and $p \rightarrow -p'$:

$$\begin{aligned}
[a_k^\dagger, a_p^\dagger] &= \frac{1}{4\pi\sqrt{\omega_k\omega_p}} [\omega_k\phi^\dagger(k) - i\pi^\dagger(k), \omega_p\phi^\dagger(p) - i\pi^\dagger(p)] = \\
&= \frac{1}{4\pi\sqrt{\omega_k\omega_p}} [\omega_k\phi(-k) - i\pi(-k), \omega_p\phi(-p) - i\pi(-p)] = \\
&= \frac{1}{4\pi\sqrt{\omega_{-k'}\omega_{-p'}}} [\omega_{-k'}\phi(k') - i\pi(k'), \omega_{-p'}\phi(p') - i\pi(p')] = \\
&= \frac{\omega_{-k'} - \omega_{-p'}}{4\pi\sqrt{\omega_{-k'}\omega_{-p'}}} 2\pi\delta(k' + p') = 0
\end{aligned} \tag{45}$$

Where we again used the fact that $\omega_k = \omega_{-k}$, this time to show that if $p' = k'$ then $\omega_{-p'} = \omega_{p'} = \omega_k$. The last commutation relation we want to check is $[a_k, a_p^\dagger]$:

$$\begin{aligned}
[a_k, a_p^\dagger] &= \frac{1}{4\pi\sqrt{\omega_k\omega_p}} [\omega_k\phi(k) + i\pi(k), \omega_p\phi^\dagger(p) - i\pi^\dagger(p)] = \\
&= \frac{1}{4\pi\sqrt{\omega_k\omega_p}} [\omega_k\phi(k) + i\pi(k), \omega_p\phi(-p) - i\pi(-p)] = \\
&= \frac{1}{4\pi\sqrt{\omega_k\omega_{-p'}}} [\omega_k\phi(k) + i\pi(k), \omega_{-p'}\phi(p') - i\pi(p')] = \\
&= \frac{1}{4\pi\sqrt{\omega_k\omega_{-p'}}} (-i\omega_k[\phi(k), \pi(p')] + i\omega_{-p'}[\pi(k), \phi(p')]) = \\
&= \frac{\omega_k + \omega_{-p'}}{4\pi\sqrt{\omega_k\omega_{-p'}}} 2\pi\delta(k+p') = \delta(k+p') = \delta(k-p)
\end{aligned} \tag{46}$$

Where in the last line, the only case when the delta function is not zero is the case where the fraction evaluates to one.

Now we will develop the Hamiltonian in terms of $a(k)$ and $a^\dagger(k)$. To do that we will add and subtract from the integrand in equation 24 the term $i\omega_k\pi(k)\phi^\dagger(k)$. Then, we will take the subtracted term and factorize only it by changing the integration variable $k \rightarrow -k$. This only moves the dagger to the $\pi(k)$ field, since $\pi^\dagger(k) = \pi(-k)$ and $\phi^\dagger(k) = \phi(-k)$, and the change of sign in the differential cancels out with the change in

order of the integration limits:

$$\begin{aligned}
H &= \frac{1}{2} \int_{-\pi}^{+\pi} \frac{dk}{2\pi} [\pi^\dagger(k)\pi(k) + \omega_k^2 \phi^\dagger(k)\phi(k)] = \\
&= \int_{-\pi}^{+\pi} \frac{dk}{4\pi} [\omega_k^2 \phi^\dagger(k)\phi(k) + \pi^\dagger(k)\pi(k) + i\omega_k (\pi(k)\phi^\dagger(k) - \pi^\dagger(k)\phi(k))] = \\
&= \int_{-\pi}^{+\pi} \frac{dk}{4\pi} [\omega_k^2 \phi^\dagger(k)\phi(k) + \pi^\dagger(k)\pi(k) + i\omega_k (\pi(k)\phi^\dagger(k) - \pi^\dagger(k)\phi(k))] = \\
&= \int_{-\pi}^{+\pi} \frac{dk}{4\pi} [\omega_k^2 \phi^\dagger(k)\phi(k) + \pi^\dagger(k)\pi(k) + i\omega_k (\phi^\dagger(k)\pi(k) - \pi^\dagger(k)\phi(k) - [\phi^\dagger(k), \pi(k)])] = \quad (47) \\
&= \int_{-\pi}^{+\pi} \frac{dk}{4\pi} [\omega_k^2 \phi^\dagger(k)\phi(k) + \pi^\dagger(k)\pi(k) + i\omega_k (\phi^\dagger(k)\pi(k) - \pi^\dagger(k)\phi(k) - [\phi(-k), \pi(k)])] = \\
&= \int_{-\pi}^{+\pi} \frac{dk}{4\pi} [\omega_k^2 \phi^\dagger(k)\phi(k) + \pi^\dagger(k)\pi(k) + i\omega_k (\phi^\dagger(k)\pi(k) - \pi^\dagger(k)\phi(k) - i2\pi\delta(-k+k))] = \\
&= \int_{-\pi}^{+\pi} \frac{dk}{4\pi} [\omega_k^2 \phi^\dagger(k)\phi(k) + i\omega_k \phi^\dagger(k)\pi(k) - i\omega_k \pi^\dagger(k)\phi(k) + \pi^\dagger(k)\pi(k) + 2\pi\omega_k\delta(0)]
\end{aligned}$$

The first four terms can be factorized:

$$\begin{aligned}
&\int_{-\pi}^{+\pi} \frac{dk}{4\pi} [(\omega_k \phi^\dagger(k) - i\pi^\dagger(k)) (\omega_k \phi(k) + i\pi(k)) + 2\pi\omega_k\delta(0)] = \\
&= \int_{-\pi}^{+\pi} dk \omega_k \left[\frac{1}{4\pi\omega_k} (\omega_k \phi^\dagger(k) - i\pi^\dagger(k)) (\omega_k \phi(k) + i\pi(k)) + \frac{1}{2}\delta(0) \right] = \quad (48) \\
&= \int_{-\pi}^{+\pi} dk \omega_k \left[a^\dagger(k)a(k) + \frac{1}{2}\delta(0) \right]
\end{aligned}$$

3.4 The Vaccum Energy relation to N

In this situation we switch the formula for the commutator $[\phi(k), \pi(p)]$:

$$[\phi(k), \pi(p)] = \sum_{-N}^N e^{-in(k+p)} \quad (49)$$

And for the situation of the vaccum energy, we got

$$[\phi(-k), \pi(k)] = \sum_{-N}^N e^{-in(-k+k)} = \sum_{-N}^N 1 = 2N + 1 \quad (50)$$

Meaning that the energy is linear to the number of points.